

MAT 2007 Introduction to Programming
The Influence Of Study Techniques on Grades

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Introduction

Research has found that study techniques that involve active recalling of information such as flash cards or self-testing, lead to better retention than passive techniques such as rereading or highlighting (Dunlosky et al., 2013). The majority of research on study techniques comes from controlled experiments, but it's unclear if the same can be concluded in regards to self-reported study habits of real students in STEM. This project investigates whether active or passive study techniques are associated with high grades among biology students. As a secondary aim, we looked at which individual study techniques are associated with higher grades. Since the data is self-reported, the analysis can show correlations, but not causational effects.

Data sample

The two datasets (Rodriguez, 2018a,b) for this project have been sourced from Harvard Dataverse (Rodriguez et al., 2018), they contain survey responses from 1326 biology students in year 1 and year 2 of their top 3 most used study techniques and their final semester grades. (scale of 1-13, the highest being the best grade). The 11 students that did not choose 3 study techniques were removed from the data set, leading to a total sample of $N = 1{,}315$. In order to answer the research question the study techniques have been grouped into active (self-testing, flash-cards, recopying from memory, condensing notes, studying with friends, diagrams) and passive (rereading, highlighting, videos, recopying notes, cramming).The category "other" was not classified. For each student a score was calculated:

$$\text{Active-passive score} = \text{No. of active techniques} - \text{No. of passive techniques.}$$

The score ranges from -3 (all passive) to $+3$ (all active), meaning a higher score indicates the student uses more active study techniques. The "other" technique is equal to 0. The primary statistical analysis was a multiple linear regression that predicts grade from the score, with cohort as a covariate. Figure 1 also includes results of pairwise Welch tests with Holm-adjusted p-values, to compare the groups. The analysis was done in Python (version 3.14.7) in Visual Studio Code. The Pandas and Statsmodels libraries were used for data processing and statistical analyses. The plots were created using Matplotlib.

Results and discussion

Based on the active–passive score, students used more active than passive techniques (mean score = 0.33, range -3 to $+3$), and most scored $+1$ (47.0%) or -1 (33.1%). A higher active–passive score was associated with higher grades after controlling for cohort ($b = 0.23$, 95% CI [0.14, 0.33], $p < .001$; Figure 1). Replacing one passive technique with an active one raises the score by 2, corresponding to a predicted grade approximately 0.47 points higher, roughly half a grade step. Cohort was not significant ($p = .271$), and the model explained only 1.8% of grade variation ($R^2 = .018$).

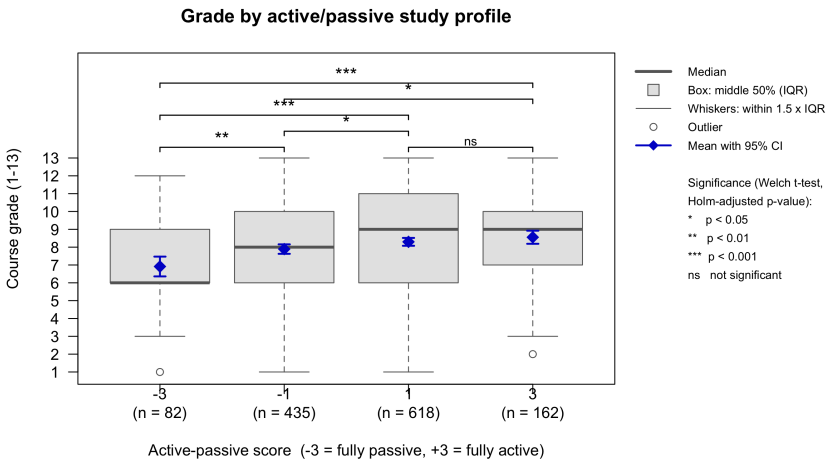


Figure 1: *Course Grade by Active–Passive Score.* Course grades (1–13) of first- and second-year biology students by active–passive score. Scores 0 and +2 are not shown because they had only 12 and 6 students.

Self-testing (65%) and rereading (56%) were the most frequently selected techniques. Figure 2 shows unadjusted mean grades and 95% confidence intervals for all 12 techniques. Because each student chose three techniques, a student appears in three groups, so the groups overlap.

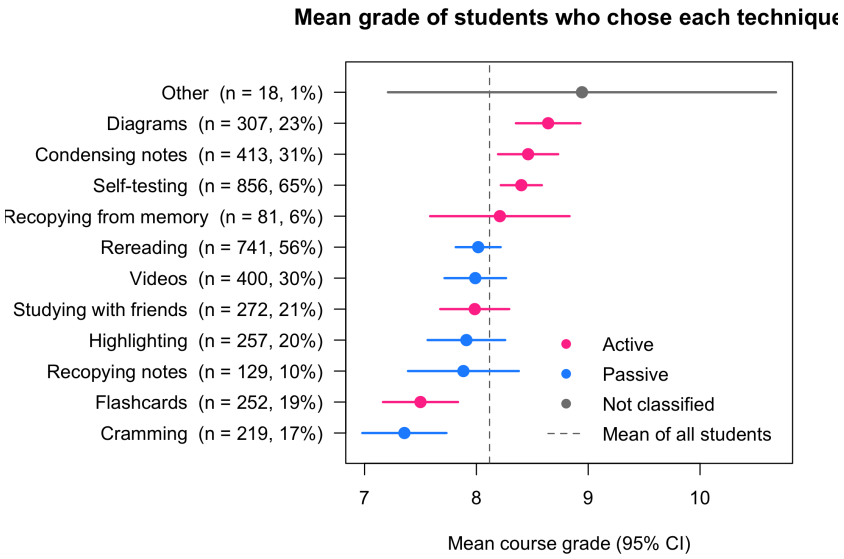


Figure 2: *Mean Grade of Students Who Chose Each Study Technique.* Mean course grades (1–13) with 95% confidence intervals for students who selected each technique ($N = 1{,}315$). The number of students selecting each technique is shown as n. Means are not adjusted for other techniques or cohorts.

Conclusion

From the statistical analysis, students who reported using a more active mix of techniques tended to obtain slightly higher grades. The survey is observational and self-reported; therefore, these findings show associations rather than causal effects. It cannot be established that active techniques cause higher grades.

Bibliography

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest, 14*(1), 4–58. https://doi.org/10.1177/1529100612453266.

Rodriguez, F., Rivas, M. J., Matsumura, L. H., Warschauer, M., & Sato, B. K. (2018). How do students study in STEM courses? Findings from a light-touch intervention and its relevance for underrepresented students. *PLOS ONE, 13*(7), e0200767. https://doi.org/10.1371/journal.pone.0200767.

Rodriguez, F. (2018). Year 1 Data [Data set]. Harvard Dataverse, V1. `https://doi.org/10.7910/DVN/LPYNOS`.

Rodriguez, F. (2018). Year 2 Data [Data set]. Harvard Dataverse, V2. `https://doi.org/10.7910/DVN/ZNWR3F`.